Session 7 · Deck 7.A1

The Learning Problem: a Handwritten Digit, and the Neuron

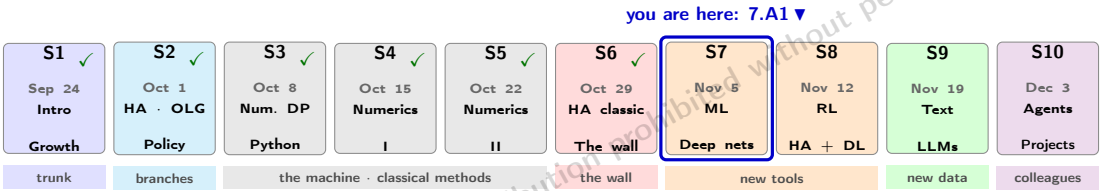
Zhigang Feng

Thursday, November 5, 2026

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Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are



Build it on paper → solve it on the machine → push it to the frontier with AI

Block A — learning a function: 7.A1 the learning problem and the neuron · 7.A2 activations and approximation · 7.A3 backpropagation and training. **Block B — using it:** 7.B1 PyTorch and a forecast · 7.B2 the growth model · 7.B3 architecture and shape · 7.B4 capacity and order.

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Roadmap: Three Parts

Sessions 3 to 6 solved models on grids with a basis chosen in advance. This deck starts one step earlier: with a problem no formula solves.

Part	Topic	The question it answers
1	Session 7, and where it starts	What will you be able to do, and what do you already have?
2	A handwritten digit	What is a learning problem, and why can it be solved at all?
3	The neuron	What is the smallest unit that learns, and what can it not do?

Idea

A learning problem is data, a family of functions f_θ, a loss, and a rule for judging the result. A neuron is the smallest member of the family: a logit.

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1 · What You Will Be Able to Do After Session 7

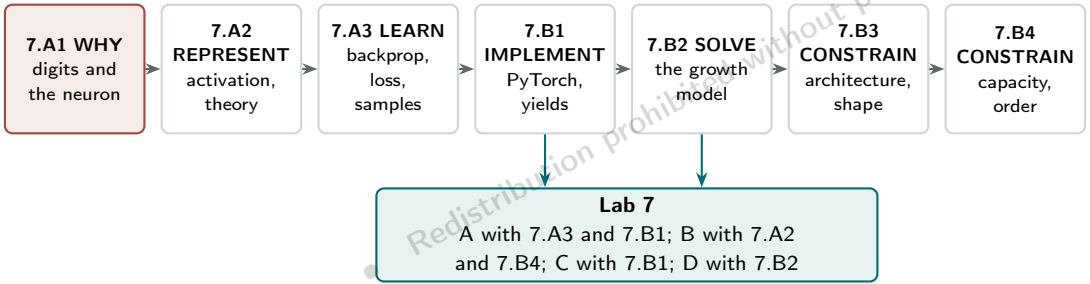
Six skills, in the order the seven decks build them.

- **Read a network as a function**: write $f_\theta(x)$, count its parameters, and say what its architecture assumes about the input (7.A1, 7.A2, 7.B3, 7.B4).
- **Check one training step by hand against autograd**: forward, loss, backward, update (7.A3, 7.B1).
- **Write the PyTorch training loop** and check it on a problem whose answer you already know (7.B1).
- **Run an honest forecast comparison**: splits by date, one information set, a random-walk baseline (7.B1).
- **Solve a growth model with a network**, and show with held-out tests that the solution is right (7.B2).
- **Throughout: a small training loss is a claim about the training sample, and nothing more.**

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1 · Session 7 in One Map

Seven decks: why, represent, learn; then implement, solve, constrain twice.



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- The running example for 7.A1 to 7.A3 is one picture: a handwritten digit.
- The economic applications are in 7.B1 (a forecast) and 7.B2 (a model solved).
- **One habit in all seven decks: test a method first where the answer is known.**

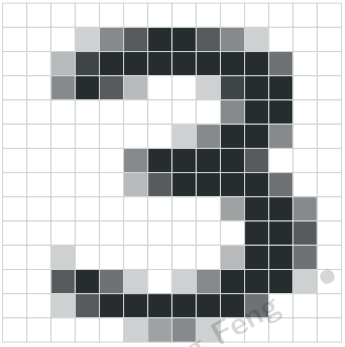
1 · What Sessions 1–6 Gave Us, and What Is New Here

The machine you already have, and the three words this session adds to it.

- **Four verbs** for what AI does in economics: measure, predict, identify, compute. Session 7 is about *predict* and *compute*; 1.A3 called the second "solving models we could not solve".
- **Three kinds of learning**: supervised (inputs with labels), unsupervised (inputs only), reinforcement (learning by acting). Session 7 is supervised; reinforcement is Session 8, motivated in 1.B3.
- **Empirical risk minimisation**: choose θ to minimise the average loss $\frac{1}{n}\sum_i \ell\big(f_\theta(x_i), y_i\big)$ over a sample — 4.A2's "empirical risk". OLS is the linear, squared-loss case.
- The rule that makes this honest: judge a model on data it never saw. 7.A3 turns it into a procedure, and every result after it is reported that way.
- 5.B1 left a promise: the projection basis need not be fixed in advance. 7.A2 and 7.B2 keep it.
- **Today starts one step earlier: with a problem no formula solves.**

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One picture carries the ideas of 7.A1 to 7.A3.

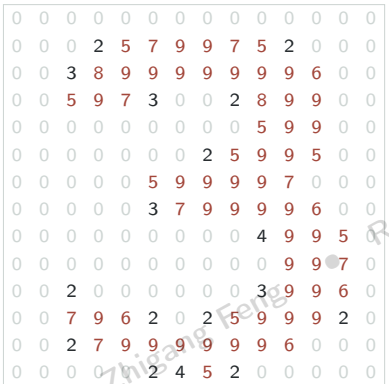


- You read this as a "3" at once, and you cannot say how.
- The benchmark is **MNIST**: 70,000 digits written by US Census Bureau staff and high-school students, each a 28×28 grey-scale image with its label.
- Drawn here at 14×14, so that every pixel can be seen.
- **Nobody writes that function down. It is learned from labelled examples.**

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Y. LeCun, L. Bottou, Y. Bengio, P. Haffner, "Gradient-based learning applied to document recognition", *Proc. IEEE* 86(11), 1998.

To the machine the picture is a table of numbers, and the table is a vector.

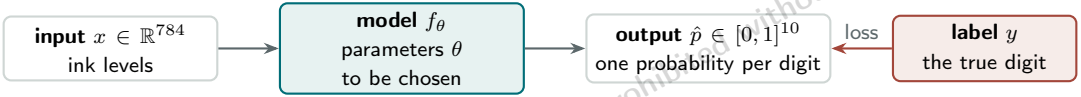


- To a computer the image is a table of ink levels: 0 is blank paper, 9 is full ink (0 to 255 in the original files).
- Read row after row, the table is a vector: 196 numbers here, $28 \times 28 = 784$ in MNIST. That vector is the entire input.
- Flattening forgets which pixels are neighbours. 7.B3 puts that back.
- **An image is a point in a space with one dimension per pixel: 196 here, 784 in MNIST.**

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The Learning Problem: from 784 Numbers to Ten Classes

Data, a model, a loss, and learning: the econometrician's problem with a richer f_θ.



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- **Data:** 60,000 labelled pairs (x_i, y_i) for training, 10,000 held out for testing.
- **Model:** a family of functions f_θ indexed by a parameter vector θ.
- **Loss:** small when $\hat{p}$ puts high probability on the true digit.
- **Learning:** choose θ to make the average training loss small, then measure on the held-out images.
- **The econometrician's problem with a richer f_θ: a specification, a criterion, an estimator.**

Try Writing the Rule for a "3"

When the rule cannot be written down, give examples and let it be estimated.



four people, one digit

- A first rule: "two open curves, stacked". Now define *curve*, *open* and *stacked* in ink levels.
- Slant, stroke width, size and position change all 784 numbers and none changes the digit.
- **Polanyi's paradox**: we know more than we can tell.
- **When the rule cannot be written, give examples and let the rule be estimated.**

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M. Polanyi, *The Tacit Dimension*, 1966; D. Autor, "Polanyi's paradox and the shape of employment growth", NBER WP 20485, 2014.

Why It Is Possible at All: Structure

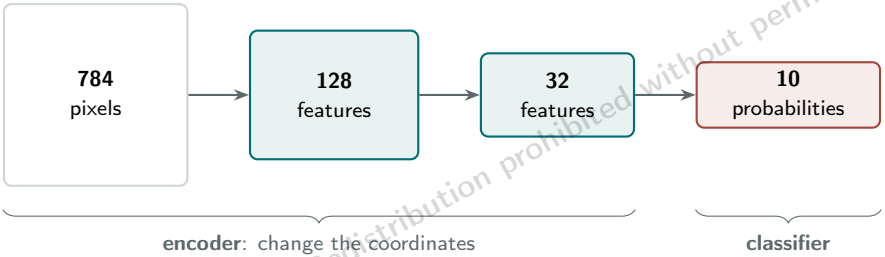
Real data occupy a small, structured part of the space they are recorded in.

- There are 256^{784} possible 28×28 images. Almost all of them look like television static.
- Images of digits are a vanishing fraction of that space, and they are not scattered at random: strokes are continuous, connected and few.
- The **manifold hypothesis**: real high-dimensional data lie near a surface of much lower dimension inside the space they are recorded in.
- Macroeconomists use the same idea. Krusell and Smith (1998) replace an infinite-dimensional wealth distribution by its mean, because in equilibrium the distribution moves in very few directions (6.B1).
- **Learning in high dimension works when the data have low-dimensional structure, and fails when they do not.**

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P. Krusell and A. Smith, "Income and wealth heterogeneity in the macroeconomy", *JPE* 106(5), 1998.

A deep network is a logit whose explanatory variables are learned.

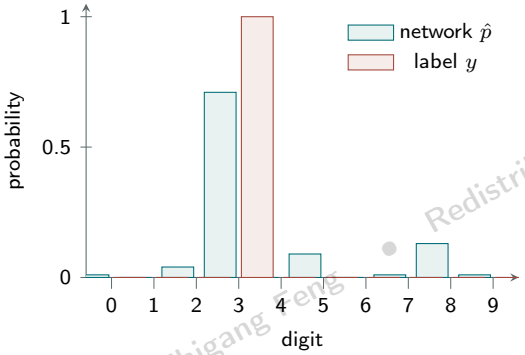


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- Each layer re-describes the input in fewer, more useful numbers. The last layer is an ordinary multinomial logit on those numbers.
- The encoder and the classifier are estimated *jointly*, by the same criterion.
- In econometric terms: the regressors are estimated, not chosen by the researcher.
- **A deep network is a logit whose explanatory variables are learned.**

Labels, One-Hot Targets, and a Probability for Each Digit

Classification is maximum likelihood for a discrete-choice model.



An illustrative $\hat{p}$, not a trained network's output.

- The label "3" is a **one-hot** vector: a 1 in position 3, zeros elsewhere.
- The **softmax** turns ten scores z_j into probabilities, $\hat{p}_j = e^{z_j} / \sum_k e^{z_k}$: McFadden's multinomial logit, with scores as utilities.
- The loss is the negative log-likelihood of the true digit: $-\log 0.71 = 0.34$ here.
- **Classification is maximum likelihood for a discrete-choice model.**

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An architecture is an assumption about the input, written into the function.

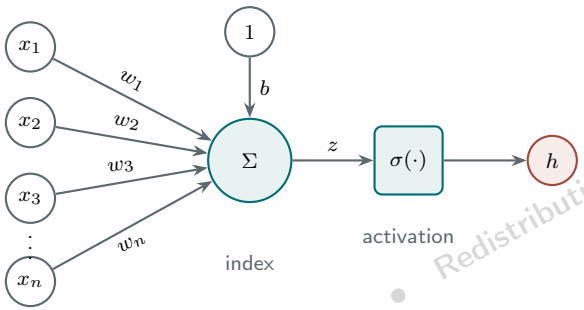
Input	Structure it carries	Architecture	Economic example
image	neighbours matter	convolutional	night lights
sequence	order matters	recurrent, transformer	policy statements
set, network	order is irrelevant	graph network	households
state vector	none assumed	dense	(k, z) in a growth model

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- 7.A1 to 7.B2 use the dense network, which assumes nothing. 7.B3 and 7.B4 return to the other rows.
- **An architecture is an assumption about the input, written into the function.**

A Neuron Is a Regression with a Nonlinear Output

A linear index, then a fixed nonlinear function: a generalised linear model.

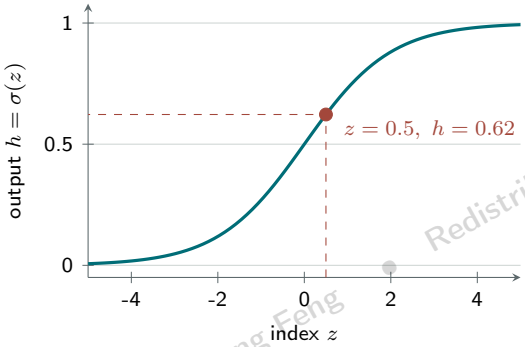


- Step 1, a linear index: $z = w^\top x + b$. The **weights** w are slopes; the **bias** b is the intercept.
- Step 2, a fixed nonlinear function, the **activation**: $h = \sigma(z)$.
- With $\sigma(z) = 1/(1 + e^{-z})$ this is exactly a logit model.
- The parameters are $\theta = (w, b)$: $n + 1$ numbers.
- **One neuron is a generalised linear model. Nothing more.**

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Miao & Feng, *AI for Economists*, §9.1.

Three inputs, three weights, one bias: the neuron is "62% on".

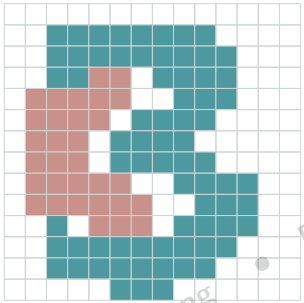


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- Inputs $x = (1.0,\ 0.5,\ 0.0)$; weights $w = (0.8,\ -0.4,\ 0.3)$; bias $b = -0.1$.
- Index: $z = 0.8(1.0) - 0.4(0.5) + 0.3(0.0) - 0.1 = 0.5$.
- Output: $h = \sigma(0.5) = 1/(1 + e^{-0.5}) = 0.62$.
- Read it as a probability: the neuron is "62% on".
- The third input is zero, so w_3 played no part. **A weight matters only where its input is active.**

One neuron asks one question of the input: how much do you look like my template?

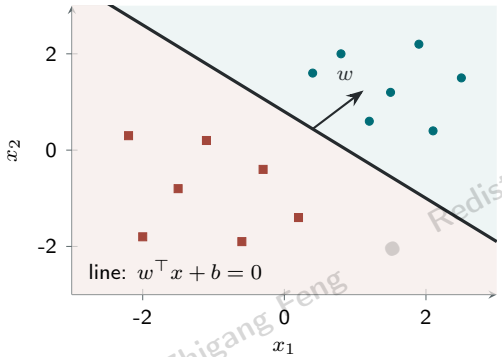


teal: positive weight **red**: negative weight

- A neuron reading the image has one weight per pixel, so its weights can be drawn as an image.
- $w^\top x$ is large when ink falls on positive weights and stays off negative ones.
- This one rewards ink where a "3" has strokes and penalises ink on the left, where an "8" closes its loops.
- **One neuron asks one question of the input: how much do you look like my template?**

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The neuron switches on in a half-space; its edge is a straight line.



- $\sigma(z) > 0.5$ exactly when $z = w^\top x + b > 0$.
- The neuron switches on in a half-space whose edge is the **hyperplane** $w^\top x + b = 0$: a line in two dimensions.
- w is perpendicular to the line; b slides it. The activation sets how sharply the output changes across the line, and does not bend it.
- **One neuron can separate only what a single straight cut can separate.**

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An Economist Has Met This Before: Logit

Same unit, same criterion, different names. What is new is stacking.

- **Binary choice.** $\Pr(y = 1 \mid x) = \sigma(w^\top x + b)$ is the logit model; $w^\top x + b$ is the difference in systematic utility.
- **Estimation.** Maximum likelihood on the logit is minimising **cross-entropy**, the loss used to train classifiers. Same criterion, different name.
- **Vocabulary.** weights = coefficients; bias = intercept; activation = link function; training = estimation; inference = prediction.
- **What is new** is not the unit. It is stacking many units so that the regressors of the last one are themselves estimated.
- **Everything you know about specification, identification and out-of-sample testing still applies.**

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D. McFadden, "Conditional logit analysis of qualitative choice behavior", in *Frontiers in Econometrics*, 1974.

Takeaways

- **Part 1.** Session 7 is supervised learning, for two jobs: predict (7.B1) and compute (7.B2). A small training loss is a claim about the training sample, and nothing more.
- **Part 2.** A learning problem is data, a family f_θ, a loss, and a rule for judging the result. An image is a vector, and learning works because real data occupy a small, structured part of the space.
- **Part 3.** A neuron is a logit: a linear index through a fixed nonlinearity, $h = \sigma(w^\top x + b)$. Alone it draws one line, and one line is not enough for the digits or for XOR.

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Bridge: The Piece We Have Not Looked At

The whole of a network's flexibility sits in that σ. We have used it without looking at it.

Take it away and a stack of layers collapses into one regression. Keep it and a network with one hidden layer can reach any continuous function on a bounded domain — which is a promise about existence, not about finding it.

Next: 7.A2

What σ is, why a layer without it buys nothing, and what a network with it can reach: XOR exactly, $\sin x$ out of hinges, and the theorems that say how far.

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